



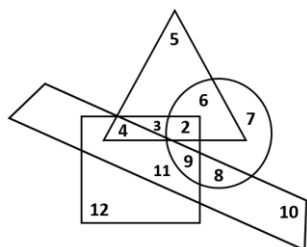
ANALYTICAL ABILITY AND LOGICAL REASONING

Directions : Q. 1-3 : Read the information given below and answer the questions that follow :

Four persons A, B, C and D play a cards game. They put Rs. 500 as stake money. When the game is over 'C' receives Rs. 19 more than 'D' and 'B' receives Rs. 21 less than 'A' whose amount was Rs. 2 less than the quarter of Rs. 500

- How much money did 'C' get ?
(a) Rs. 147 (b) Rs. 136
(c) Rs. 144 (d) Rs. 159
- How much money did 'B' get ?
(a) Rs. 102 (b) Rs. 107
(c) Rs. 108 (d) Rs. 110
- Who get highest amount ?
(a) A (b) B (c) C (d) D

Directions : Q. 4- 6 : In the following diagram circle stands for 'educated', square for 'hardworking', triangle for 'urban people' and rectangle for 'honest'. Different regions in the diagram are numbered from 2 to 12. Study the diagram carefully and answer.



- Educated, hard-working and urban people are indicated by
(a) 7 (b) 2 (c) 3 (d) 4
- Non-urban educated people who are neither hardworking nor honest are indicated by
(a) 5 (b) 7 (c) 10 (d) 12
- Honest, educated and hardworking non urban people are indicated by
(a) 3 (b) 4 (c) 6 (d) 9
- Mr. X left his entire estate to his wife, his daughter, his son and the cook. His daughter and son got half the estate, sharing in the ratio of 4 to 3. His wife got twice as much as the son. If the cook received a bequest of Rs.500, then the entire estate was
(a) Rs. 3,500 (b) Rs. 5,500
(c) Rs. 6,500 (d) Rs. 7,000
- At a dance party a group of girls and boys exchange dances as follows :
One boy dances with 5 girls, Second boy dances with 6 girls, and so on last boy dances with all girls. If b represents the number of boys and g represents the number of girls, then
(a) $b = g$ (b) $b = g / 5$
(c) $b = g - 4$ (d) $b = g - 5$
- The average age of husband and wife was 22 years when they were married five years back. What is the present

average age of the family if they have a three year old child?

- (a) 19 Years (b) 25 Years
(c) 27 Years (d) $28\frac{1}{2}$ Years

10. Which of the following will be acceptable for establishing a fact?

- (a) Opinion of large number of people
(b) Traditionally in practice over a long period of time
(c) Availability of observable evidences
(d) References in the ancient literature

Directions : Q. 11-14 : Six scientists A, B, C, D, E and F are to present at paper each at a one-day conference. Three of them will present their papers in the morning session before the lunch break whereas the other three will be presented in the afternoon session. The lectures have to be scheduled in such a way that they comply with the following restrictions:

B should present his paper immediately before C's presentation; their presentations cannot be separated by the lunch break. D must be either the first or the last scientist to present his paper.

11. In case C is to be the fifth scientist to present his paper, then B must be

- (a) first (b) second
(c) third (d) fourth

12. B could be placed for any of the following places in the order of presenters EXCEPT

- (a) second (b) third
(c) fourth (d) fifth

13. In case F is to present his paper immediately after D presents his paper, C's could be scheduled for which of the following places in the order of presenters?

- (a) second (b) third
(c) fourth (d) fifth

14. In case F and E are the fifth and sixth presenters respectively then which of the following must be true?

- (a) A is first in the order of presenters
(b) A is third in the order of presenters
(c) A is fourth in the order of presenters
(d) B is first in the order of presenters

Directions : Q. 15 -18 : Study the following information to answer the given questions :

- In a family of 6 persons, there are two couples.
- The lawyer is the head of the family and has only two sons- Mukesh and Rakesh-both teachers.
- Mrs. Reena and her mother-in-law both are lawyers.
- Mukesh's wife is a doctor and they have a son, Ajay.

15. What is the profession of Rakesh's wife?

- (a) Teacher (b) Doctor
(c) Lawyer (d) None of these

16. How many male members are there in the family?

- (a) Two (b) Three
(c) Four (d) None of these

17. What is/was Ajay's grandfather's occupation?

- (a) Teacher (b) Lawyer





- (c) Doctor (d) Cannot be determined
18. What is the profession of Ajay?
- (a) Teacher (b) Lawyer
- (c) Doctor (d) Cannot be determined

Directions : Q. 19 - 21 :

Mrs. Thomes received a large order for stitching school uniforms from Mayflower school and Littleflower school. She has two cutters who will cut the fabric, five tailors who will do the stitching and two assistants to stitch the button and button holes. Each of these nine persons will work for exactly 10 hours a day. Each of the Mayflower uniforms requires 20 min. for cutting the fabric, one hour for stitching, and 15 min. for stitching buttons and button holes, whereas the Littleflower uniform requires 30 min., 1 hour and 30 min. respectively for these activities.

19. What is the number of Littleflower uniforms that Mrs. Thomes can complete in a day ?
- (a) 50 (b) 20 (c) 40 (d) 30
20. On a particular day, Mrs. Thomes decided to complete 20 Littleflower uniforms. How many Mayflower uniforms can she complete on that day ?
- (a) 30 (b) 40 (c) 20 (d) 0
21. If she hires one more assistant, what is the maximum number of Mayflower uniforms that she can complete in a day ?
- (a) 40 (b) 50 (c) 60 (d) 30
22. In a certain code, RIPPLE is written as 613382 and LIFE is written as 8192. How is PILLER written in that code?
- (a) 318826 (b) 318286 (c) 618826 (d) 328816
23. A doctor said to his compounder "I go to see the patients at their residence after every 3:30 hours. I have already gone to the patient 1:20 hours ago and next time I shall go at 1.40 pm". At what time this information was given to the compounder by the doctor ?
- (a) 10.10 a.m. (b) 11.30 a.m.
- (c) 11.20 a.m. (d) None of these

Directions : Q. 24 : In the following question three statements are followed by a conclusion. Study the statements and the conclusion and point out which statement studied together will bring to the conclusion

24. Statements:

- (i) Price rise is natural phenomenon
- (ii) If production increases prices fall
- (iii) High prices affect the poor

Conclusion : If production rises the poor feel relieved. Answer Choices

- (a) Only i and ii (b) Only i and iii
- (c) Only ii and iii (d) Data Insufficient

25. In how many different ways can the letters of the word "DETAIL" be arranged in such a way that the vowels occupy only the odd positions?

- (a) 32 (b) 36 (c) 48 (d) 60

26. If from 4 to 55 the numbers which are divisible by 3 and the numbers which contain 3 as one of the digits, are removed, then how many numbers will be left ?

- (a) 24 (b) 23 (c) 22 (d) 25

27. In the following number-series, one term is wrong. Which term is wrong ?

5, 12, 19, 33, 47, 75, 104

- (a) 33 (b) 47 (c) 75 (d) 104

28. The position of A in a class is 5th from the top and position of B is 7th from the bottom. If C is at 6th place after A and 6th place before B, how many students are there in the class ?

- (a) 25 (b) 23 (c) 21 (d) 22

29. Suppose $X = 2^{100}$, $Y = 3^{100}$ and $Z = 4^{100}$. Exactly one of the following is true. Which is it?

- (a) $X + Y = Z$ (b) $X + Y < Z$
- (c) $X + Y > Z$ (d) $XY = Z$

30. Assume that the following three statements are true.

- (A) All freshmen are human
- (B) All students are human
- (C) Some students think

Given the following four statements :

- (I) All freshmen are students
- (II) Some humans think
- (III) No freshmen think
- (IV) Some humans who think are not students

Those which are logical consequences of I, II and III are

- (a) 2 (b) 4 (c) 2, 3 (d) 1, 2

31. Five persons A, B, C, D and E were travelling in a car. There were two ladies in the group. Two knew car driving, of them one was a lady. A is brother of D. B, wife of D drove at the beginning, E drove at the end. Who was the other lady in the group?

- (a) D (b) B (c) C (d) E

32. Choose which pair of numbers carries next in the following sequence :

61, 57, 50, 61, 43, 36, 61

- (a) 29, 61 (b) 27, 20
- (c) 31, 61 (d) 29, 22

33. Correct the following equation by inter-changing two signs. $3 - 9 \times 27 + 9 \div 3 = 3$

- (a) + and - (b) $\times$ and +
- (c) $\times$ and $\div$ (d) $\times$ and -

34. Pushpa is twice as old as Rita was two years ago. If the difference between their ages be 2 years, how old is Pushpa today ?

- (a) 6 years (b) 8 years
- (c) 10 years (d) 12 years

35. A clock is set right at 8 a.m. The clock gains 10 minutes in 24 Hrs. What will be the right time when the clock indicates 1 p.m. on the following day ?

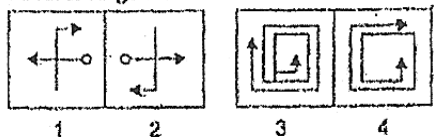
- (a) 11.40 p.m. (b) 12.48 p.m.
- (c) 12 noon (d) 10 p.m.



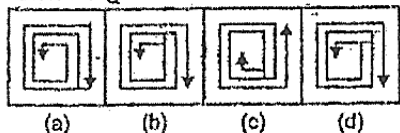


36. Choose the best answer figure to substitute element 4 in the problem figures so that element 3 is related to element 4 in the same way as element 1 is related to element 2.

Problem Figures



Answer Figures



Directions : Q. 37 : In the following question below are given two statements followed by four conclusions numbered I, II, III, IV. You have to take the two given statements to be true even if they seem to be at variance from commonly known facts. Read all the conclusions and then decide which of the given conclusions logically follows from the two given statements, disregarding commonly known facts.

37. Statements :

- A) Some green are blue
- B) No blue is white
- Conclusions
- (I) Some blue are green
- (II) Some white are green
- (III) Some green are not white
- (IV) All white are green

- (a) Only I follows
- (b) Only II and III follows
- (c) Only I and III follows
- (d) Only I and II follows

Directions : Q. 38-40 : In each of the 3 questions below, are given four statements followed by four conclusions numbered I, II, III, IV. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. Read all the conclusions and then decide which of the given conclusions logically follows from the given statements disregarding commonly known facts.

38. Statements : Some doctors are lawyers. All teachers are lawyers. Some engineers are lawyers. All engineers are businessman.

Conclusions :

- I) Some teachers are doctors.
- II) Some businessmen are lawyers.
- III) Some businessmen are teachers.
- IV) Some lawyers are teachers.

- (a) None follows
- (b) Only II follows
- (c) Only III follows
- (d) Only II and IV follow

39. Statements : All plastics are glasses. Some sponges are glasses. All sponges are clothes. All clothes are liquids.

Conclusions :

- I) All liquids are sponges.
- II) Some plastics are clothes.
- III) All glasses are plastics.
- IV) All liquids are clothes.

- (a) None follows

(b) Only either II or IV follows

(c) Only III and IV follows

(d) Only I and IV follows

40. Statements : All sands are beaches. All shores are beaches. Some beaches are trees. All trees are hotels.

Conclusions :

I) Some shores are hotels.

II) All beaches are shores.

III) Some beaches are hotels.

IV) Some sands are trees.

(a) Only III follows

(b) Only II follows

(c) Only IV follows

(d) None of these

ANSWER

1.	A	2.	A	3.	C	4.	B	5.	B
6.	d	7.	d	8.	c	9.	a	10.	c
11.	d	12.	b	13.	D	14.	C	15.	C
16.	B	17.	B	18.	D	19.	C	20.	A
21.	B	22.	A	23.	B	24.	C	25.	B
26.	D	27.	D	28.	B	29.	B	30.	A
31.	C	32.	D	33.	D	34.	B	35.	B
36.	b	37.	c	38.	d	39.	a	40.	a





SOLUTION

- (a) C's Amount = Rs. 147.
- (a) B's Amount = Rs. 102.
- (c) C get the highest amount Rs. 147.
- (b) No. 2 represent the educated, hardworking and urban people.
- (b) No. 7 represents non-urban educated people who are neither hard working nor honest.
- (d) Number 9 represents honest, educated and hardworking non-urban people.
- (d) Let the entire estate was Rs. x .
(Daughter + Son)'s estate = Rs. $\frac{x}{2}$
 $\therefore$ Daughter and son sharing ratio is 4 : 3
 $\therefore$ Daughter's estate = Rs. $\frac{x}{2} \times \frac{4}{7} = \text{Rs. } \frac{4x}{14}$
Son's estate = Rs. $\frac{x}{2} \times \frac{3}{7} = \text{Rs. } \frac{3x}{14}$
So, mother's estate = $2 \times \text{Rs. } \frac{3x}{14} = \text{Rs. } \frac{6x}{14}$
Remaining estate = $x - \left(\frac{x}{2} + \frac{6x}{14}\right) = x - \frac{13x}{14} = \text{Rs. } \frac{x}{14}$
Then, $\frac{x}{14} = 500 \Rightarrow x = \text{Rs. } 7000$
- (c) Suppose number of boys = b
1st boy dances with 5 girls ($1 + 4 = 5$)
2nd boy dances with 6 girls ($2 + 4 = 6$)
3rd boy dances with 7 girls ($3 + 4 = 7$)
 $\therefore$ bth boy dances with $(b + 4)$ girls
Hence, $g = b + 4 \Rightarrow b = g - 4$
- (a) Five years ago total age of husband and wife
 $22 \times 2 = 44\text{yr}$
Present age of husband and wife = $44 + 2 \times 5$
 $44 + 10 = 54\text{ yr}$
 $\therefore$ Total age of the family = $54 + 3 = 57\text{ yr}$
Hence, average age of the family = $\frac{57}{3} = 19\text{yr}$
- (c) Availability of observable evidences will be acceptable for establishing a fact.

Solutions [Q. Nos. 11 to 14]

From the given information, we can definitely say that B's presentation will be immediately before C's presentation and their presentation together will be either pre lunch or post lunch. D is either first or last scientist to present his paper.

- (d) If C is fifth scientist to present his paper, then B must be fourth to present his paper because B's presentation will be immediately before C's presentation.
- (b) We know that B and C scientists presentation cannot be separated by the lunch break, in this case B could be placed in third order of presenters.
- (d) From the given information, we can say that, D is the first scientist and F is the second scientist, then C can be present in only either fifth or sixth place. As B and C have to present in two consecutive position either pre lunch or post lunch.
- (c) If F and E are fifth and sixth scientists, then D will be the first scientist. Band C will be the second and third presenters. Hence, A is the fourth presenter.

Solutions [Q. Nos. 15 to 18]

Persons	Occupation	Male/Female	Relationship
Grandmother	Lawyer	Female	Reena's mother-in-law
Mukesh	Teacher	Male	
Rakesh	Teacher	Male	
Reena	Lawyer	Female	Rakesh's wife
Ajay	_____	Male	Mukesh's son
Mukesh's wife	Doctor	Female	

- (c) Rakesh's wife profession is a lawyer.
- (b) Three male members in the family.
- (b) We cannot determined Ajay's grandfather occupation.
- (d) We cannot determined the profession of Ajay.

Solutions (Q. Nos. 19-21)

Time taken for stitching school uniforms

School	Cutting	Stitching	Button Stitching
May flower	20 min	60 min	15 min
Little flower	30 min	60 min	30 min

Number of fabric cutters = 2

Number of tailors = 5

Number of assistants = 2

Each of them do the work for 10 h.

Time duration for fabric cutters = $2 \times 10\text{ h}$

= 20 h

= 1200 min

Time duration for tailors

= $5 \times 10\text{ h}$

= 50 h

= 3000 min

Time duration for button stitchers = $2 \times 10\text{ h}$

= 20 h

= 1200 min

Maximum number of Little flower uniforms that can be cut in a day = $\frac{3000}{30} = 40$

Maximum number of Little flower uniforms that can be stitch in a day = $\frac{3000}{60} = 50$

Maximum number of Little flower uniforms that can be buttoned in a day = $\frac{1200}{30} = 40$

- (c) Number of Little flower uniforms that Mrs Thomas can complete in a day = 40

- (a) If Mrs Thomas decided to complete 20 Little flower uniforms.

Then, time duration for fabric cutters

= $1200 - (20 \times 30)$

= 600 min

Time duration for tailors = $3000 - (20 \times 60) = 1800\text{ min}$

Time duration for assistants = $1200 - (20 \times 30)$

= 600 min

Hence, number of May flower - uniforms that can be completed in a day

= minimum of $\left(\frac{600}{20}, \frac{1800}{60}, \frac{600}{15}\right)$

= minimum of (30, 30, 40) = 30





21. (b) If Mrs Thomas hires one more assistants, then time duration for button stitchers

$$= 1200 + (1 \times 10 \times 60) = 1800 \text{ min}$$

Hence, maximum May flower uniforms that can be completed in a day

$$= \text{minimum of } \left(\frac{1200}{20}, \frac{3000}{60}, \frac{1800}{15} \right)$$

$$= \text{minimum of } (60, 50, 120) = 50$$

22. (a)

R	I	P	P	L	E		L
↓	↓	↓	↓	↓	↓		↓
6	1	3	3	8	2	And	8

P	I	L	L	E	R
↓	↓	↓	↓	↓	↓
3	1	8	8	2	6

Hence,

23. (b) Doctor go to see the patients before

$$1:40 \text{ pm} = 1:40 \text{ pm} - 3:30$$

$$= 13:40 \text{ pm} - 3:30 \text{ h} = 10:10 \text{ am}$$

$$\text{Hence, required time} = 10:10 + 1:20 = 11:30 \text{ am}$$

24. (c) Only statements (ii) and (iii) logically follow the conclusion.

25. (b)

E		A			
1	2	3	4	5	6

$$\text{Hence, number of ways} = 3! \times 3! = 3 \times 2 \times 3 \times 2 = 36$$

26. (d) Numbers between 4 to 55 which are divisible by 3 and contain 3 as one of the digits are 6, 9, 12, 13, 15, 18, 21, 23, 24, 27, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 42, 43, 45, 48, 51, 53, 54 = 27 numbers.

$$\text{Hence, remaining number} = 52 - 27 = 25$$

27. (d) The pattern of the series is

5	12	19	33	47	75	103	104
	+7	+7	+14	+14	+28	+28	

Hence, 104 is the wrong number.

28. (b) C's position from the top = 5th 6th 11th

$$\text{C's position from the bottom} = 7\text{th} + 6\text{th}$$

$$= 13\text{th } 5\text{th}$$

A Hence, total number of students in the class

$$= (11+13) - 1$$

$$= 23 \text{ 6th}$$

29. (b) Suppose, $2^2 + 3^2 < 4^2 (\because 4 + 9 < 16)$

$$\Rightarrow 2^3 + 3^3 < 4^3 (\because 8 + 27 < 64)$$

$$\Rightarrow 2^4 + 3^4 < 4^4 (\because 16 + 81 < 256)$$

So, solving the pattern be conclude that for higher value

$$n > 1$$

$$2^n + 3^n < 4^n$$

$$\text{Then, } X + Y < Z$$

30. (a) According to the statements, Venn-diagram is

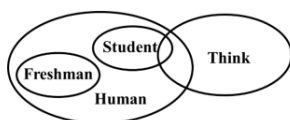
Conclusions

I. ×

II. ✓

III. ×

IV. ×



31. (c) $A^+ \xleftarrow{\text{Brother}} D^+ \xrightarrow{\text{Wife}} B - (\text{Drove})$
 $\left[\begin{array}{l} + \rightarrow \text{male} \\ - \rightarrow \text{Female} \end{array} \right]$
 $E^+ (\text{Drove})$

Hence, C is definitely the other lady in the group.

32. (d) The given pattern follows the three series.

$$\text{I} \rightarrow 61 \xrightarrow{+0} 61 \xrightarrow{+0} 61$$

$$\text{II} \rightarrow 57 \xrightarrow{-14} 43 \xrightarrow{-14} 29$$

$$\text{III} \rightarrow 50 \xrightarrow{-14} 36 \xrightarrow{-14} 22$$

33. (d) Given equation, $3 - 9 \times 27 + 9 \div 3 = 3$

From option (d), interchanging the signs

$$3 \times 9 - 27 + 9 + 3 = 3$$

$$\Rightarrow 27 - 27 + 3 = 3$$

$$\Rightarrow 3 = 3$$

34. (b) Let the present age of Pushpa be x yr. Then, the present age of Rita would be (x - 2) yr. So, $x = 2(x - 2 - 2)$

$$\Rightarrow x = 2x - 8$$

$$\Rightarrow x = 8 \text{ yr}$$

Therefore, the present age of Pushpa is 8 yr.

35. (b) Time from 8 am of a particular day to 1 pm on the following day = 29 h. Now, the clock gains 10 min in 24 h, it means that 24 h 10 min of this clock is equal to 24 h of the correct clock.

$$145 \text{ h of this clock} = 24 \text{ h of the correct clock } \frac{145}{6} \text{ h of this clock} = 24 \text{ h of the correct clock}$$

$$\frac{24}{145} \times 6 \times 29 = 28 \text{ h } 48 \text{ min of the correct clock.}$$

It means that the clock in question is 12 min faster than the correct clock. Therefore, when clock indicates 1 pm the correct time will be 48 min past 12.

36. (b) Answer figure (b) is correct substitute of the problem figure (4).

37. (c) According to the statements, Venn-diagram is

Conclusions

I. ✓

II. ×

III. ✓

IV. ×



38. (d) According to the statements, Venn-diagram is

Conclusions

I. ×

II. ✓

III. ×

IV. ✓



39. (a) According to the statements, Venn-diagram is

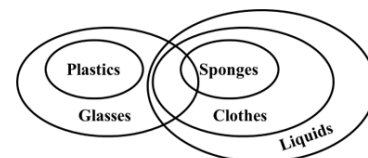
Conclusions

I. ×

II. ×

III. ×

IV. ×



40. (a) According to the statements, Venn-diagram is

Conclusion

I. ×

II. ×

III. ✓

IV. ×

